

Homework #4

Entropy Rate and Data Compression

Introduction to Information Theory 2026 · EC5203 / EC4310 / MM4450 · Prof. Heung-No Lee · GIST

How to submit. Put your solution in the `submission/` folder of your assignment repository: a single PDF (`submission/hw4.pdf` — scanned handwriting is fine if legible), or Markdown with L^AT_EX (`submission/hw4.md`). Commit and push before the deadline (23:59 KST); the last commit before the deadline is graded. If you worked as a group (max 3), list all members at the top — each member still submits in their own repository.

Questions. Light ones in the AI TA chat on the course homepage; rigorous derivations and proofs on the course GitHub Q&A (mention @claude). Before the deadline you get hints, never solutions.

Problem HW4.1 (C&T 5.12) Shannon codes and Huffman codes. Consider a random variable X that takes on four values with probabilities $\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{4}, \frac{1}{12}\right)$.

- (a) Construct a Huffman code for this random variable.
- (b) Show that there exist two different sets of optimal lengths for the codewords; namely, show that codeword length assignments $(1, 2, 3, 3)$ and $(2, 2, 2, 2)$ are both optimal.
- (c) Conclude that there are optimal codes with codeword lengths for some symbols that exceed the Shannon code length

$$\left\lceil \log \frac{1}{p(x)} \right\rceil.$$

Problem HW4.2 (C&T 5.16) Huffman codes. Consider a random variable X that takes six values $\{A, B, C, D, E, F\}$ with probabilities 0.5, 0.25, 0.1, 0.05, 0.05, and 0.05, respectively.

- (a) Construct a binary Huffman code for this random variable. What is its average length?
- (b) Construct a quaternary Huffman code for this random variable [i.e., a code over an alphabet of four symbols (call them a, b, c and d)]. What is the average length of this code?
- (c) One way to construct a binary code for the random variable is to start with a quaternary code and convert the symbols into binary using the mapping $a \rightarrow 00$, $b \rightarrow 01$, $c \rightarrow 10$, and $d \rightarrow 11$. What is the average length of the binary code for the random variable above constructed by this process?
- (d) For any random variable X , let L_H be the average length of the binary Huffman code for the random variable, and let L_{QB} be the average length code constructed by first building a quaternary Huffman code and converting it to binary. Show that

$$L_H \leq L_{QB} < L_H + 2. \tag{5.146}$$

- (e) The lower bound in the example is tight. Give an example where the code constructed by converting an optimal quaternary code is also the optimal binary code.
- (f) The upper bound (i.e., $L_{QB} < L_H + 2$) is not tight. In fact, a better bound is $L_{QB} \leq L_H + 1$. Prove this bound, and provide an example where this bound is tight.

Problem HW4.3 (C&T 5.18) *Classes of codes.* Consider the code $\{0, 01\}$.

- (a) Is it instantaneous?
- (b) Is it uniquely decodable?
- (c) Is it nonsingular?

Problem HW4.4 (C&T 5.20) *Huffman codes with costs.* Words such as “Run!”, “Help!”, and “Fire!” are short, not because they are used frequently, but perhaps because time is precious in the situations in which these words are required. Suppose that $X = i$ with probability p_i , $i = 1, 2, \dots, m$. Let l_i be the number of binary symbols in the codeword associated with $X = i$, and let c_i denote the cost per letter of the codeword when $X = i$. Thus, the average cost C of the description of X is $C = \sum_{i=1}^m p_i c_i l_i$.

- (a) Minimize C over all $l_1, l_2, \dots, l_m$ such that $\sum 2^{-l_i} \leq 1$. Ignore any implied integer constraints on l_i . Exhibit the minimizing $l_1^*, l_2^*, \dots, l_m^*$ and the associated minimum value C^* .
- (b) How would you use the Huffman code procedure to minimize C over all uniquely decodable codes? Let C_{Huffman} denote this minimum.
- (c) Can you show that

$$C^* \leq C_{\text{Huffman}} \leq C^* + \sum_{i=1}^m p_i c_i ?$$

Problem HW4.5 (C&T 5.21) *Conditions for unique decodability.* Prove that a code C is uniquely decodable if (and only if) the extension

$$C^k(x_1, x_2, \dots, x_k) = C(x_1)C(x_2) \cdots C(x_k)$$

is a one-to-one mapping from $\mathcal{X}^k$ to D^* for every $k \geq 1$. (The “only if” part is obvious.)

Problem HW4.6 (C&T 4.4) *Second law of thermodynamics.* Let $X_1, X_2, X_3, \dots$ be a stationary first-order Markov chain. In Section 4.4 it was shown that $H(X_n | X_1) \geq H(X_{n-1} | X_1)$ for $n = 2, 3, \dots$. Thus, conditional uncertainty about the future grows with time. This is true although the unconditional uncertainty $H(X_n)$ remains constant. However, show by example that $H(X_n | X_1 = x_1)$ does not necessarily grow with n for every x_1 .

Problem HW4.7 (C&T 4.7) *Entropy rates of Markov chains.*

- (a) Find the entropy rate of the two-state Markov chain with transition matrix

$$P = \begin{bmatrix} 1 - p_{01} & p_{01} \\ p_{10} & 1 - p_{10} \end{bmatrix}.$$

- (b) What values of p_{01}, p_{10} maximize the entropy rate?
- (c) Find the entropy rate of the two-state Markov chain with transition matrix

$$P = \begin{bmatrix} 1 - p & p \\ 1 & 0 \end{bmatrix}.$$

- (d) Find the maximum value of the entropy rate of the Markov chain of part (c). We expect that the maximizing value of p should be less than $\frac{1}{2}$, since the 0 state permits more information to be generated than the 1 state.
- (e) Let $N(t)$ be the number of allowable state sequences of length t for the Markov chain of part (c). Find $N(t)$ and calculate

$$H_0 = \lim_{t \rightarrow \infty} \frac{1}{t} \log N(t).$$

[Hint: Find a linear recurrence that expresses $N(t)$ in terms of $N(t-1)$ and $N(t-2)$. Why is H_0 an upper bound on the entropy rate of the Markov chain? Compare H_0 with the maximum entropy found in part (d).]

Problem HW4.8 (C&T 4.24) Entropy rates. Let $\{X_i\}$ be a stationary process. Let $Y_i = (X_i, X_{i+1})$. Let $Z_i = (X_{2i}, X_{2i+1})$. Let $V_i = X_{2i}$. Consider the entropy rates $H(\mathcal{X})$, $H(\mathcal{Y})$, $H(\mathcal{Z})$, and $H(\mathcal{V})$ of the processes $\{X_i\}$, $\{Y_i\}$, $\{Z_i\}$, and $\{V_i\}$. What is the inequality relationship $\leq$, $=$, or $\geq$ between each of the pairs listed below?

- (a) $H(\mathcal{X}) \leq H(\mathcal{Y})$.
- (b) $H(\mathcal{X}) \leq H(\mathcal{Z})$.
- (c) $H(\mathcal{X}) \leq H(\mathcal{V})$.
- (d) $H(\mathcal{Z}) \leq H(\mathcal{X})$.

Problem HW4.9 (C&T 5.6) Bad codes. Which of these codes cannot be Huffman codes for any probability assignment?

- (a) $\{0, 10, 11\}$
- (b) $\{00, 01, 10, 110\}$
- (c) $\{01, 10\}$

Problem HW4.10 (C&T 5.7) Huffman 20 questions. Consider a set of n objects. Let $X_i = 1$ or 0 accordingly as the i th object is good or defective. Let $X_1, X_2, \dots, X_n$ be independent with $P\{X_i = 1\} = p_i$; and $p_1 > p_2 > \dots > p_n > \frac{1}{2}$. We are asked to determine the set of all defective objects. Any yes–no question you can think of is admissible.

- (a) Give a good lower bound on the minimum average number of questions required.
- (b) If the longest sequence of questions is required by nature's answers to our questions, what (in words) is the last question we should ask? What two sets are we distinguishing with this question? Assume a compact (minimum average length) sequence of questions.
- (c) Give an upper bound (within one question) on the minimum average number of questions required.

Problem HW4.11 Lempel–Ziv on real data; optional. Use the provided MATLAB file on Lempel–Ziv coding and compress each file (`Jordan_text.txt` and `H1N1_virus.txt`) and report the compression ratio. Use the provided H1N1 virus gene: extract the virus gene part from `H1N1_virus.txt`, i.e., `atgaatccaa accaaaagat...` Calculate the entropy of the gene. Do the LZ coding of the gene. Comment on the efficiency of the LZ code, compared with the entropy of the gene.